

Power Shield:

An Automated Yield-Loss Alerting (AYLA) Pipeline Using Explainable AI (XAI) and Machine Learning (ML) for Power Grid Attack Detection and LLM-Guided Remediation

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Abstract

Modern power grids face an escalating threat from coordinated cyber-physical attacks, including False Data Injection Attacks (FDIA), Relay Saturation and Current Attacks (RSCA), and Remote Terminal and Communication Injection (RTCI), which are designed to evade conventional anomaly detectors by mimicking plausible physical behavior. While machine learning classifiers can flag anomalies with high recall, they offer no actionable guidance to operators—a critical gap in real-world deployments where response speed and accuracy are paramount. This paper presents **Power Shield**, implementing an **Automated Yield-Loss Alerting (AYLA)** pipeline—a four-stage explainable AI (XAI) system that addresses this gap end-to-end and proves its feasibility statistically. A SHAP-guided LightGBM/XGBoost stacked ensemble first scores each grid event, achieving **91.88%** accuracy and **94.6%** attack recall on a held-out test set of 15,676 samples drawn from a 78,377-record SCADA dataset. SHAP TreeExplainer then identifies the top contributing sensors, which are used as semantically enriched retrieval keys against a ChromaDB vector store indexed from thirteen operator-written grid-protocol manuals. Candidates are reranked by a cross-encoder before being passed to a locally hosted Mistral-Nemo LLM that reasons via a structured Chain-of-Thought (CoT) step before generating step-by-step remediation advice. Critically, Stages 2–4 (SHAP, RAG, and LLM) activate only when Stage 1 flags an event as a suspected attack or degradation—not on every incoming sensor snapshot—so the computationally intensive LLM inference is incurred only on the small fraction of readings that genuinely require operator explanation, making the pipeline practical for real-time SCADA deployment on commodity hardware. An end-to-end evaluation demonstrates that the full AYLA pipeline is feasible and applicable to operational ICS security.

1 Introduction

Power grids are among the most critical infrastructure systems in modern society. Disruptions—whether caused by natural faults or deliberate attacks—can cascade into widespread outages affecting hospitals, emergency services, and millions of civilians. The Ukrainian blackouts of 2015 and 2016, attributed to state-sponsored cyber actors, demonstrated that adversaries are capable of weaponizing grid control software with devastating physical consequences [1]. Since then, the threat landscape has only intensified, with industrial control system (ICS) attacks increasing by over 140% between 2020 and 2024.

Traditional Supervisory Control and Data Acquisition (SCADA) security relies on rule-based intrusion detection systems (IDS) tuned to known attack signatures. These systems fail against novel or stealthy attacks, particularly *False Data Injection Attacks* (FDIA)—a class of cyber-

physical attack in which an adversary manipulates sensor telemetry to mislead state estimation algorithms without triggering threshold-based alarms [2]. Machine learning (ML) classifiers have emerged as a promising complement to signature-based systems, offering the ability to learn complex, multi-variate anomaly patterns directly from sensor data [3].

However, detection alone is insufficient. An operator who receives an alert without an explanation of which sensors triggered it, which attack scenario it matches, and what corrective steps to take is at best slowed and at worst misled. Current commercial ICS/OT security platforms surface alerts but do not generate automated, grounded, step-by-step remediation guidance. The field of Explainable AI (XAI) has produced methods—most notably SHAP [4]—that can attribute a classifier’s decision to individual input features, but these attributions are expressed as raw numerical weights that require expert interpretation. More

recently, Retrieval-Augmented Generation (RAG) [5] and Large Language Models (LLMs) have demonstrated the ability to synthesize structured knowledge with natural-language generation, opening a path toward fully automated, human-readable incident response.

Prior work achieving high detection accuracy on the MSU/ORNL power grid dataset [6–8] evaluates classifiers on individual attack sub-scenarios in isolation, eliminating the inter-class ambiguity that characterises combined-dataset deployment. The only prior work to apply explainability to the combined MSU/ORNL dataset [9] demonstrates SHAP-based feature attribution but does not extend to automated protocol retrieval or operator remediation—the operational gap that motivates AYLA.

Power Shield implements an **Automated Yield-Loss Alerting (AYLA)** pipeline that unifies these advances into a single end-to-end system. Its principal contributions are:

- C1. SHAP-guided feature selection and RAG query construction.** SHAP global importance first reduces 128 sensor features to a 29-feature subset, improving classifier generalization. At inference time, SHAP attributions identify the top-contributing sensors and enrich their names with semantic type descriptors (e.g., R3-PM2:V $\rightarrow$ R3-PM2:V (voltage magnitude)) before forming the retrieval query, improving embedding alignment with protocol language. Prior work applies SHAP to grid security and intrusion detection for post-hoc feature explanation [9, 10]; AYLA instead uses per-event SHAP attributions as the *retrieval query* submitted to the vector store, directly linking the classifier’s explanation layer to document retrieval. To the author’s knowledge, this use of per-event SHAP attributions as semantic retrieval keys is novel.
- C2. Cross-encoder reranking for transparent protocol disambiguation (XAI).** Bi-encoder multi-pass retrieval first discovers both relay-specific and generic protocol candidates; a cross-encoder then rescores each candidate jointly against the primary query, producing relevance signals directly comparable across all retrieval passes and eliminating secondary-query score pollution without sacrificing generic-protocol recall. The ranked candidate list with explicit cross-encoder scores constitutes an XAI artifact: for each alert, operators and auditors can inspect exactly which protocol documents were considered, their relative relevance scores, and why the top-ranked document was selected—providing a transparent, traceable decision trail that supports post-hoc review without relying on black-box inference.
- C3. Chain-of-Thought protocol selection.** The LLM prompt structures the response to require an explicit reasoning step that checks the RANK #1 retrieved protocol’s diagnostic signature against the observed

sensor pattern before committing to a diagnosis, yielding measurably higher protocol attribution accuracy than a passive retrieval hint.

- C4. Ablation-confirmed LLM grounding in retrieved context.** A no-context ablation withholds retrieved protocol documents from the LLM prompt while keeping all other inputs—SHAP drivers, sensor readings, classifier verdict—identical. Attribution accuracy drops from 90.5% to **16.5%** (33/200, 95% CI [12.0%, 22.3%])—a **74.0 percentage-point decline**—confirming that the full-pipeline accuracy reflects genuine grounding in retrieved documents rather than parametric knowledge. A χ^2 goodness-of-fit test ($\chi^2 = 1931.42$, $p < 10^{-10}$) further confirms performance far exceeds the random-selection baseline ($p_0 = 1/13 \approx 0.077$).
- C5. Comparison with real-world deployed ICS security systems.** AYLA is compared against current commercial ICS/OT platforms (Claroty, Dragos, Nozomi Networks) and NERC CIP standards, demonstrating that the pipeline provides automated, grounded, step-by-step remediation guidance—a capability absent from all existing deployed systems.

2 Background and Related Work

2.1 Power Grid Cyber-Physical Attacks

The power grid presents a uniquely complex attack surface because physical constraints (Kirchhoff’s laws, protection relay logic) interact with cyber systems (SCADA, EMS, PMUs) in ways that sophisticated adversaries can exploit. FDIA, introduced formally by Liu et al. [2], constructs a vector $\mathbf{a}$ such that $\mathbf{z}' = \mathbf{z} + \mathbf{a} = H\hat{\mathbf{x}} + \mathbf{c}$ where $\mathbf{z}'$ is the corrupted measurement, H is the system measurement matrix, $\hat{\mathbf{x}}$ is the estimated state, and $\mathbf{c}$ is a consistent attack vector that passes the bad-data detection residual test. Related attack classes in the dataset include: *RSCA* (Relay Saturation and Current Attack), which manipulates harmonic content in impedance and current phase angle measurements; *RTCI* (Remote Terminal Communication Injection), which induces current collapse correlated with IDS log spikes; and *FREQ* attacks, which perturb frequency across all relay zones simultaneously.

2.2 Dataset Generation

The MSU/ORNL ICS Security Dataset [11] was generated using a hardware-in-the-loop power grid simulation testbed at Mississippi State University. The physical testbed consisted of a real-time digital simulator running a 4-relay transmission protection system model, communicating via DNP3 and Modbus protocols over a simulated SCADA network. Attack scenarios were injected by researchers who crafted each attack class to target specific relay telemetry channels while bypassing the threshold-based bad-data

detection residual test. Each of the 15 CSV files in the binary dataset corresponds to one distinct experimental run, combining one or more attack scenarios with normal grid operation. The binary label (Attack/Natural) is assigned by the testbed’s ground-truth event logger, not by any detector. Natural rows include both nominal operation and benign fault events (e.g., load switching, phase faults), making the classification boundary more challenging than a pure attack-vs-nominal formulation.

2.3 Real-World ICS Security Context

Current deployed ICS/OT security platforms—including Claroty, Dragos, and Nozomi Networks—primarily provide passive network monitoring and signature-based anomaly detection. They surface alerts but do not generate automated, grounded, step-by-step remediation guidance. NERC CIP (Critical Infrastructure Protection) standards mandate monitoring and incident logging but do not require automated response generation, leaving operators to manually consult voluminous protocol documentation during live incidents. AYLA addresses this gap by providing not only detection but also LLM-generated, protocol-grounded remediation guidance within the same pipeline.

2.4 Machine Learning for ICS Intrusion Detection

Gradient-boosted decision trees have become the dominant ML paradigm for tabular ICS sensor data due to their robustness to non-stationarity, interpretability, and handling of class imbalance. Stacking ensembles—where a meta-learner combines the probabilistic outputs of multiple base learners—have consistently outperformed single-model approaches on heterogeneous tabular benchmarks [6].

A critical methodological distinction separates the prior literature on this dataset: whether evaluation is performed on individual per-scenario sub-datasets or the combined heterogeneous dataset. Hink et al. [12], the foundational ML study on the MSU/ORNL data, trained a Random Forest on the full combined dataset and reported 90.56% accuracy—the baseline against which this work directly competes. Subsequent work using individual sub-datasets reports substantially higher figures: Panthi and Das [6] achieved 98.63%; Naeem et al. [7] reported 95.82%; and a graph-attention network [8] reached 97.63%. These results are *not directly comparable* to combined-dataset evaluation: isolating a single attack scenario eliminates inter-class ambiguity, dramatically simplifying the classification boundary. The only prior work known to the author that applies explainability to the combined MSU/ORNL data is [9] (Craveiro et al.), who use Permutation SHAP to explain ML predictions and report above 96% on individual scenarios but do not extend to protocol attribution or operator remediation. AYLA operates on the full combined dataset and surpasses the Hink et al. baseline while adding SHAP attribution, RAG-based protocol retrieval,

and LLM-guided step-by-step remediation—capabilities absent from all prior work on this dataset.

2.5 Explainable AI for Cybersecurity

SHAP (SHapley Additive exPlanations) [4] provides theoretically grounded feature attributions by computing Shapley values from cooperative game theory. TreeSHAP [13] computes exact SHAP values for tree ensembles in polynomial time, making it practical for real-time attribution. SHAP has been applied to network intrusion detection [10], but its use as a *retrieval key* for RAG systems is, to the author’s knowledge, novel.

2.6 Retrieval-Augmented Generation and the IHGR-RAG Framework

RAG [5] conditions an LLM on dynamically retrieved context, enabling factual, grounded responses without full model retraining. IHGR-RAG [14] addresses semantically overlapping guideline retrieval in power equipment condition assessment via hierarchical retrieval and LLM-based Chain-of-Thought reranking. The AYLA pipeline directly adopts the cross-encoder reranking and CoT paradigm from IHGR-RAG [14] and extends it to the SCADA relay security domain with SHAP-attribution-guided query construction. The key design decisions and their motivating sources are detailed in Appendix A.

2.7 LLMs for Decision Support

Instruction-tuned LLMs have demonstrated strong zero-shot performance on structured reasoning tasks when provided with sufficient context. Chain-of-Thought (CoT) prompting [15] consistently improves accuracy on tasks requiring multi-step logic. Mistral-Nemo [16], a 12B-parameter model with a 128k context window, provides the balance of context capacity and inference speed appropriate for real-time grid operations on commodity hardware.

3 System Architecture

The AYLA pipeline processes each grid event through four sequential stages. Stages 1 and 2 operate on raw sensor data; Stages 3 and 4 operate on the symbolic output of Stage 2. A key design property is that Stages 2–4 (SHAP attribution, RAG retrieval, and LLM remediation) activate only when Stage 1 exceeds the alert threshold τ : the classifier runs at machine speed on every incoming sensor snapshot, while the computationally intensive LLM inference is reserved for the small fraction of events that actually require operator explanation and remediation guidance. This event-driven design is what makes the pipeline viable for real-time deployment without dedicated high-throughput inference infrastructure.

Stage 1 — Probabilistic Classification. A LightGBM/XGBoost stacked ensemble, trained on 29 SHAP-selected features from an initial pool of 128, ingests a sensor snapshot and outputs $P(\text{Attack})$ via a LogisticRegression meta-learner. If this probability exceeds the operating threshold $\tau = 0.560$, the event is flagged and passed to Stage 2.

Stage 2 — SHAP Attribution. TreeSHAP computes exact SHAP values ϕ_i for all 128 features via the LightGBM base model. The top- k positive contributors (sensors pushing toward Attack) are selected as the *alert drivers*. Log channels (SCADA, relay, and Snort IDS logs) are separated from physical sensor drivers and reported independently. Each sensor name is expanded to a semantic descriptor (e.g., `R1-PA:ZH` $\rightarrow$ `R1-PA:ZH (impedance harmonic)`) using a relay-and-type regex registry.

Stage 3 — RAG Protocol Retrieval. A two-phase retrieval strategy queries a ChromaDB vector store indexed from thirteen operator-written grid-protocol manuals using the `all-MiniLM-L6-v2` sentence embedding model. *Phase 1* (bi-encoder, multi-pass): a relay-filtered primary query ($k=4$) surfaces relay-specific FDIA protocols; supplementary type-pattern queries ($k=2$) surface generic protocols not named by relay sensor identifiers. *Phase 2* (cross-encoder reranking, adopted from IHGR-RAG): all unique candidates are scored jointly against the primary query by a `ms-marco-MiniLM-L-6-v2` cross-encoder. When frequency sensors (`:F`, `:DF`) dominate the alert drivers, a supplementary frequency-deviation query is scored and its cross-encoder scores are summed with the primary scores, boosting `FREQ-01` without penalizing relay-specific FDIA cases.

Stage 4 — LLM Remediation. The cross-encoder-ranked protocol context, SHAP attribution summary, raw sensor readings, and classifier verdict are assembled into a structured prompt. A CoT reasoning step instructs the LLM to explicitly verify the RANK #1 protocol’s diagnostic signature against the sensor readings before committing to a diagnosis. The model then outputs: (1) Protocol Match Reasoning, (2) Diagnosis with protocol ID and sensor explanation, and (3) numbered remediation steps drawn directly from the matched manual.

4 Mathematical Formulation

The AYLA pipeline’s three novel computational components are formalized here; full derivations for the gradient-boosting objective, Shapley value formulation, evaluation metrics, and Wilson confidence intervals are provided in Appendix B.

SHAP-guided feature selection. An initial LightGBM model trained on all $d=128$ features computes global mean absolute SHAP importance:

$$\bar{\phi}_j = \frac{1}{N_{\text{sel}}} \sum_{i=1}^{N_{\text{sel}}} |\phi_j^{(i)}| \quad (1)$$

on a held-out sample of $N_{\text{sel}} = 3000$ training rows. Features with $\bar{\phi}_j \geq \frac{1}{d} \sum_{k=1}^d \bar{\phi}_k$ are retained, reducing $d=128$ to $d'=29$ features.

Ensemble stacking. A LogisticRegression meta-learner h combines LightGBM and XGBoost attack probabilities on a held-out calibration set:

$$P(\text{Attack} | \mathbf{x}) = \sigma(\beta_0 + \beta_1 P_{\text{LGB}}(\mathbf{x}_S) + \beta_2 P_{\text{XGB}}(\mathbf{x}_S)) \quad (2)$$

where $\mathbf{x}_S$ is the SHAP-selected feature subset, $\beta_1 = 6.74$, $\beta_2 = 0.68$, and $\tau = 0.560$ is selected by sweeping $[0.30, 0.90]$ subject to false alert rate $< 15\%$.

Cross-encoder reranking. The cross-encoder f_{CE} scores each query-document pair jointly:

$$\hat{s}_j = f_{\text{CE}}(q, d_j) + f_{\text{CE}}(q_{\text{freq}}, d_j) \quad (3)$$

where q_{freq} is added only when frequency sensors dominate the alert drivers. Unlike bi-encoder L2 scores, cross-encoder scores are directly comparable across all retrieval passes regardless of which pass found each candidate.

5 Dataset and Experimental Setup

5.1 Dataset

Experiments use the Mississippi State University / Oak Ridge National Laboratory (MSU/ORNL) ICS Security Dataset [11], a widely used benchmark for power system intrusion detection research. The binary variant used here contains **78,377 samples** with **128 features** covering four simulated protection relays (R1–R4). Features include:

- **Power meter measurements:** voltage magnitudes (`PM1:V–PM12:V`), current magnitudes (`PM4:I–PM12:I`)
- **Phase angle measurements:** impedance (`PA:Z`, `PA:ZH`), voltage phase harmonics (`PA1:VH–PA6:VH`), current phase harmonics (`PA1:IH–PA12:IH`)
- **Relay-level measurements:** frequency (`Rx:F`), frequency rate-of-change (`Rx:DF`), apparent power (`Rx:S`)
- **Log channels:** `control_panel_log1--4`, `relay1--4_log`, `snort_log1--4`

The class distribution is 55,663 Attack (71.0%) and 22,714 Natural (29.0%), yielding a 2.45:1 imbalance ratio corrected via `is_unbalance=True` (LightGBM) and `scale_pos_weight` (XGBoost). Data is split 64/16/20 into training, calibration, and held-out test sets. The 20% test set contains 15,676 samples and is never used during training or calibration.

5.2 Protocol Knowledge Base

Thirteen operator-written protocol documents are stored in a single structured text file and indexed into ChromaDB using 700-character chunks with 100-character overlap. Documents are tagged with relay-specific metadata

(R1–R4) or **GENERIC** based on their protocol header, enabling relay-filtered retrieval. The thirteen protocols cover: four relay-specific FDIA scenarios (R1-FDIA, R2-FDIA, R3-FDIA, R4-PM), three generic anomaly classes (RSCA-01, RTCI-01, FREQ-01), four fault/degradation scenarios (CURR-01, PHASE-01, FAULT-01, DEGR-01, DEGR-02), and one normal-operation baseline (EXPECTED-09).

5.3 Implementation

The AYLA pipeline is implemented in Python across five modular scripts: `preprocess_raw_data.py`, `train_classifier_v2.py`, `build_rag_index.py`, `grid_monitor.py`, and `run_once.py`, with shared model classes in `model_utils.py`. Table 1 summarizes the component configuration. The full implementation, pre-trained model, and evaluation data are publicly available at <https://github.com/gradams42/power-shield>. Detailed per-module implementation notes are provided in Appendix C.

5.4 Evaluation Protocol

Classifier evaluation is performed on the full 15,676-sample test set.

RAG retrieval accuracy is evaluated using 13 curated hand-crafted test cases (one per protocol), each specifying a relay, a set of sensor names, and the expected protocol. A test case is a “hit” if the expected protocol identifier appears anywhere in the retrieved context string.

LLM protocol attribution accuracy is evaluated on a stratified random sample of n triggered samples (equal Attack/Natural split from events above τ). For each sample, the ground-truth protocol is determined by matching the top SHAP sensors against the curated protocol sensor lists using a weighted overlap score. A response is correct if the expected protocol identifier appears anywhere in the LLM’s generated text. Protocol attribution is also tested against the random-selection null hypothesis ($p_0 = 1/13 \approx 0.077$) using a χ^2 goodness-of-fit test.

No-context ablation re-evaluates the same 200 samples with the retrieved protocol context replaced by a placeholder, keeping all other prompt inputs (SHAP drivers, sensor readings, classifier verdict) identical. The accuracy drop between conditions isolates the contribution of RAG retrieval to LLM attribution accuracy.

6 Results

6.1 Stage 1: Classifier Performance

Table 2: AYLA ensemble classifier performance on the held-out 20% test set ($n=15,676$). SE denotes standard error. CI denotes Wilson 95% confidence interval.

Metric	Value	SE	95% CI
Overall Accuracy	91.88%	0.22%	[91.44%, 92.30%]
Attack Recall	94.55%	0.22%	[94.11%, 94.95%]
Natural Recall	85.34%	0.52%	[84.28%, 86.34%]
False Alert Rate	14.66%	0.52%	[13.66%, 15.72%]
F1 (weighted)	91.86%	—	—
MCC	0.8020	—	—
Cohen’s κ	0.8020	—	—
AUC-ROC	0.9665	—	—

Table 3: Confusion matrix for AYLA ensemble classifier at $\tau=0.560$. Rows = actual class; Columns = predicted class.

	Pred: Attack	Pred: Natural
Actual: Attack	10,526	607
Actual: Natural	666	3,877

The ensemble achieves an AUC-ROC of 0.9665, indicating strong discriminative power across all operating thresholds. An MCC of 0.8020 confirms high balanced classification performance despite the 2.45:1 class imbalance. Attack recall of 94.6% at $\tau=0.560$ meets the primary safety requirement of minimizing missed attacks. The AYLA classifier’s 91.88% accuracy exceeds the Hink et al. [12] combined-dataset baseline of 90.56%. The full threshold sweep analysis is provided in Appendix E.

6.2 Stage 3: RAG Retrieval Accuracy

The AYLA RAG stage achieves perfect recall on all 13 curated test cases, confirming that the two-phase retrieval strategy reliably surfaces the correct protocol within the retrieved context. The RANK #1 column reveals the core challenge: for several protocols (R4-PM, RTCI-01, CURR-01, FREQ-01), the correct document is retrieved but does not achieve RANK #1 from the bi-encoder alone—the cross-encoder reranking step and LLM CoT reasoning must work together to surface the correct protocol.

Table 1: AYLA system component configuration.

Component	Configuration
Classifier	LightGBM + XGBoost ensemble (29/128 SHAP-selected features)
Meta-learner	LogisticRegression ($\beta_1=6.74$, $\beta_2=0.68$)
Alert threshold	$\tau = 0.560$ (threshold-sweep optimised, false alert < 15%)
SHAP	TreeExplainer (exact, LightGBM base model; zero-padded to 128-d)
Embedding model	all-MiniLM-L6-v2 (384-dim)
Vector database	ChromaDB (L2 distance)
RAG Phase 1	Bi-encoder; $k=4$ relay-filtered + $k=2$ type-pattern passes
RAG Phase 2	Cross-encoder reranking (ms-marco-MiniLM-L-6-v2)
LLM	Mistral-Nemo 12B (via Ollama, local)
LLM temperature	0.0 (deterministic)
LLM max tokens	2,048

Table 4: AYLA RAG retrieval accuracy on the 13 curated test cases.

Protocol	Relay	RANK #1 Retrieved	Hit
R1-FDIA	R1	R1-FDIA	✓
R2-FDIA	R2	R2-FDIA	✓
R3-FDIA	R3	R3-FDIA	✓
R4-PM	R4	EXPECTED-09	✓
RTCI-01	—	EXPECTED-09	✓
RSCA-01	R1	RSCA-01	✓
FREQ-01	—	R1-FDIA	✓
CURR-01	R2	EXPECTED-09	✓
PHASE-01	R3	PHASE-01	✓
FAULT-01	R2	R2-FDIA	✓
DEGR-01	R4	EXPECTED-09	✓
DEGR-02	R2	R2-FDIA	✓
EXPECTED-09	R1	R1-FDIA	✓
Overall Retrieval Accuracy			13/13 (100%)
Relay-Filtered Accuracy			11/11 (100%)

6.3 Stage 4: LLM Protocol Attribution Accuracy

Table 5: AYLA LLM protocol attribution on the 200-sample evaluation. “Any citation” denotes that at least one known protocol was cited; “Correct” denotes that the ground-truth protocol was cited. χ^2 null hypothesis: random selection ($p_0=1/13$).

Metric	Value	95% CI
Any citation rate	200/200 (100.0%)	[98.1%, 100%]
Correct citation rate	181/200 (90.5%)	[85.6%, 93.8%]
χ^2 vs. random	1931.42	$p < 10^{-10}$

The χ^2 statistic of 1931.42 ($p < 10^{-10}$) confirms full-pipeline performance far exceeds the random-selection baseline ($p_0 = 1/13$).

No-context ablation. To directly confirm that this accuracy reflects grounding in retrieved documents rather

than parametric knowledge, the same 200 samples were re-evaluated with the protocol context withheld from the LLM prompt; all other inputs remained identical.

Table 7: No-context ablation: LLM protocol attribution with and without retrieved context (same 200 samples, identical prompt structure).

Condition	Correct	95% CI
Full pipeline (w/ RAG context)	181/200 (90.5%)	[85.6%, 93.8%]
No-context ablation	33/200 (16.5%)	[12.0%, 22.3%]
Accuracy drop		−74.0 pp

Without context, attribution accuracy drops to 16.5%—a 74.0 pp decline (Table 7). The no-context LLM collapses to citing a single default protocol (R4-PM) for nearly all events regardless of relay identity or sensor type, consistent with parametric guessing rather than document-grounded reasoning. The non-overlapping 95% confidence intervals ([85.6%, 93.8%] vs. [12.0%, 22.3%]) confirm the gap is

Table 6: Protocol-level attribution accuracy from the AYLA 200-sample evaluation. Totals reflect the sample allocation across ground-truth protocols.

Protocol	Expected	Correct	Primary Error
R3-FDIA	98	96 (98.0%)	EXPECTED-09 $\times$ 1, PHASE-01 $\times$ 1
R4-PM	33	33 (100.0%)	—
R1-FDIA	33	32 (97.0%)	RSCA-01 $\times$ 1
FREQ-01	12	9 (75.0%)	R2-FDIA/R4-PM/PHASE-01 mix $\times$ 3
R2-FDIA	11	9 (81.8%)	PHASE-01 $\times$ 2
RTCI-01	7	0 (0.0%)	R4-PM $\times$ 3, PHASE-01 $\times$ 1, other $\times$ 3
CURR-01	4	1 (25.0%)	various $\times$ 3
DEGR-02	2	1 (50.0%)	R2-FDIA mix $\times$ 1
Total	200	181 (90.5%)	

statistically unambiguous.

RTCI-01 is the dominant remaining failure mode (0/7 = 0%). When the ground truth is RTCI-01, the alert sensors are typically phase angle or current harmonics—types shared with multiple FDIA and PHASE-01 protocols. These sensor types overlap more strongly with FDIA protocol text than with the Snort IDS log spikes that most clearly identify RTCI-01, causing the cross-encoder to rank relay-specific FDIA documents above RTCI-01.

A complementary cascaded per-scenario classifier providing sub-millisecond triage is described in Appendix D.

7 Architecture Comparison and Practical Trade-offs

Two benchmark runs are compared to characterize AYLA’s improvement over the original architecture and to identify practical trade-offs for deployment planning.

Run A (Original Architecture) used a single LightGBM classifier with isotonic calibration ($\tau=0.60$, accuracy 91.76%, AUC 0.9644, MCC 0.7992) and achieved 84.5% correct protocol citation (169/200, 95% CI [78.8%, 88.9%]) with any-citation rate of 198/200 (99.0%).

Run B (AYLA, Proposed) uses the LightGBM+XGBoost ensemble with cross-encoder reranking ($\tau=0.560$, accuracy 91.88%, AUC 0.9665, MCC 0.8020) and achieves 90.5% correct protocol citation (181/200, 95% CI [85.6%, 93.8%]) with any-citation rate of 200/200 (100.0%).

AYLA achieves a **6.0 percentage-point improvement** in LLM protocol attribution accuracy over the original architecture, with non-overlapping confidence intervals confirming this is a statistically meaningful gain.

Key Observation: Failure Pattern Shift

A structurally important difference between the two runs is the *character* of the remaining failures, not just the count.

In Run A, failures are concentrated and predictable: RTCI-01 always mismatched to R4-PM (both failures, 0/2), and FREQ-01 failed on every sample (0/15 = 0%). An

operator who knew these systematic failure modes could route RTCI-01 and FREQ-01 events to alternative handling without any LLM involvement.

In Run B (AYLA), FREQ-01 improves dramatically to 9/12 = 75.0%, and RTCI-01 fails across a broader variety of misattributions, making the 0/7 failure rate less predictable.

Practical implication: For deployments where RTCI-01 events are independently captured by Snort IDS threshold alerts, Run A’s concentrated and predictable failure mode is operationally more manageable. AYLA is the preferred choice when comprehensive, protocol-accurate attribution across all 13 protocols is required and no external Snort-based RTCI-01 detection is available.

8 Discussion

8.1 Remaining Failure Modes

RTCI-01’s 0/7 failure rate is the dominant unsolved problem. RTCI-01 involves current collapse correlated with Snort IDS log spikes; however, the ensemble’s SHAP attribution frequently elevates phase-angle or current-harmonic sensors above the log channels when both co-occur. When log channels do not surface as the primary SHAP driver, the RAG query lacks the vocabulary to distinguish RTCI-01 from relay-specific FDIA scenarios. A targeted fix was attempted: promoting Snort log channel drivers in the RAG query when they are active, regardless of relative SHAP rank. However, this approach introduced new attribution failures in other protocol classes—Snort log activity co-occurs with multiple attack types, so unconditionally promoting log channels over-retrieves RTCI-01 for events that are actually relay-specific FDIA or FREQ-01. RTCI-01 therefore remains an open failure class. Resolving it likely requires a fundamentally different approach—such as a dedicated log-channel detection stage that treats Snort spikes as a hard retrieval filter, or a protocol-aware query template that separates the log-channel signal from the sensor-type signal before retrieval.

FREQ-01 improved substantially with cross-encoder

Table 8: Architecture comparison: Run A (original) vs. Run B (AYLA, proposed). 95% Wilson CIs shown for LLM accuracy. †: Run A any-citation 2/200 had no protocol cited; Run B achieves 100% any-citation.

Config.	LLM Acc.	Any Cite	RTCI-01	FREQ-01	R3-FDIA	Failure Pattern
Original arch.	84.5% [78.8%, 88.9%]	198/200	0/2	0/15	88/98	Concentrated (RTCI+FREQ always fail)
AYLA (proposed)	90.5% [85.6%, 93.8%]	200/200	0/7	9/12	96/98	Distributed (RTCI only systematic)

reranking (0% $\rightarrow$ 75.0%), confirming that the fundamental barrier was score incomparability across retrieval passes rather than missing protocol coverage. The three residual FREQ-01 failures occur in samples where a relay-specific voltage sensor dominates the primary SHAP driver. Additional analysis of CURR-01, DEGR-02, and PHASE-01 failure modes is provided in Appendix F.

8.2 Hardware and Deployment Requirements

All AYLA benchmark results were produced on an NVIDIA RTX 3060 laptop GPU (6 GB VRAM), averaging **3.5 minutes per event** end-to-end. Critically, the LLM stage is event-driven: it triggers *only* when a sensor snapshot exceeds τ , so the 3.5-minute cost is incurred only on flagged anomalies rather than every sensor reading. For production deployments, an RTX 4090 reduces latency to approximately 45–90 s per event ($\sim 3\times$ bandwidth improvement); an A100 to 20–40 s ($\sim 6\times$). For single-substation deployments where attack events are rare (1–10 per day), laptop-class hardware is viable; multi-substation control centers should use a dedicated RTX 4090 or A100-class GPU server.

8.3 Relation to IHGR-RAG and Real-World Deployment

AYLA directly adopts the cross-encoder reranking and CoT paradigm from IHGR-RAG [14], validating that the CoT+cross-encoder architecture generalizes from power equipment condition assessment to cyber-physical attack attribution. The primary methodological difference is that AYLA uses SHAP-attribution-guided primary queries rather than hierarchical retrieval with global context injection.

In production, AYLA would operate as a second-tier alerting layer behind the existing SCADA IDS. Unlike Claroty, Dragos, and Nozomi Networks—which surface alerts but provide no automated remediation guidance—AYLA delivers step-by-step, protocol-grounded advice for each flagged event. The 14.7% false alert rate is competitive with industrial thresholds and adjustable via the threshold sweep (Appendix E) without retraining. The local LLM

deployment is critical for environments where SCADA telemetry cannot be transmitted to external API endpoints.

9 Conclusion

This paper presented Power Shield, implementing an end-to-end Automated Yield-Loss Alerting (AYLA) pipeline for power grid attack detection and LLM-guided remediation. Three principal results establish the pipeline’s feasibility.

First, classifier feasibility: the SHAP-guided LightGBM/XGBoost stacked ensemble achieves AUC-ROC 0.9665, MCC 0.8020, and attack recall 94.6% at $\tau=0.560$, surpassing the Hink et al. [12] combined-dataset baseline (90.56%) on the full heterogeneous MSU/ORNL dataset.

Second, LLM grounding confirmed by ablation: AYLA achieves 100% RAG retrieval accuracy across all 13 protocol test cases and 90.5% correct protocol citation (181/200, 95% CI [85.6%, 93.8%]). A no-context ablation—identical prompts with retrieved documents withheld—drops attribution accuracy to 16.5% (33/200, 95% CI [12.0%, 22.3%]), a 74.0 pp decline with non-overlapping confidence intervals that directly confirms the LLM is grounding its responses in the retrieved protocol manuals. A χ^2 test ($\chi^2 = 1931.42$, $p < 10^{-10}$) further confirms full-pipeline performance far exceeds the random-selection baseline.

Third, comparison with deployed systems: unlike current commercial ICS security platforms (Claroty, Dragos, Nozomi Networks), which provide passive network monitoring and alert surfacing but no automated remediation guidance, AYLA delivers step-by-step, protocol-grounded remediation advice for each flagged event—a capability absent from all known deployed ICS security systems.

An architecture comparison between the original single-LightGBM system (Run A, 84.5% LLM accuracy) and AYLA (Run B, 90.5%) shows a 6.0 pp improvement attributable to the ensemble classifier and cross-encoder reranking, while also revealing a shift from concentrated predictable failures to distributed failures with RTCI-01 as the only systematic remaining failure mode.

The full implementation, pre-trained model, and evaluation data are publicly available at <https://github.com/gradams42/power-shield>.

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A Architecture Design Decisions

The key AYLA design decisions and their motivating sources are:

- **LightGBM+XGBoost stacking:** motivated by Panthi and Das [6], whose metaheuristic ensemble demonstrates that complementary base learners outperform single models on MSU/ORNL data.

- **SHAP for retrieval keys:** novel extension of SHAP attribution to RAG query construction—SHAP values identify *which* sensors drove the alert, and those sensor names (with semantic descriptors) form the retrieval query, ensuring the LLM receives context grounded in the same features that triggered the classifier.
- **Cross-encoder reranking:** adopted directly from IHGR-RAG [14], which demonstrated that joint query-document scoring resolves the score incomparability problem inherent in multi-pass bi-encoder retrieval.
- **CoT prompting:** following Wei et al. [15], AYLA’s prompt requires the LLM to explicitly state the RANK #1 protocol’s diagnostic signature and verify it against the observed sensor pattern before committing to a diagnosis.

B Mathematical Formulation

Gradient Boosting Classification

LightGBM [17] learns an additive ensemble of T decision trees. At iteration t , a new tree f_t is fit to the negative gradient of the loss function $\mathcal{L}$:

$$f_t = \arg \min_f \sum_{i=1}^N [-g_i f(\mathbf{x}_i) + \frac{1}{2} h_i f(\mathbf{x}_i)^2] + \Omega(f) \quad (4)$$

where $g_i = \partial_{\hat{y}} \ell(y_i, \hat{y}_i^{(t-1)})$ and $h_i = \partial_{\hat{y}}^2 \ell(y_i, \hat{y}_i^{(t-1)})$ are the first- and second-order gradients of the per-sample loss, and $\Omega(f)$ is a leaf-weight regularization term. For binary classification, the loss is the log-loss $\ell(y, \hat{y}) = -[y \log \sigma(\hat{y}) + (1-y) \log(1-\sigma(\hat{y}))]$ where σ is the sigmoid function. The same gradient-boosting objective is used by XGBoost [18].

SHAP Shapley Values

For a trained model f with feature set $N = \{1, \dots, d\}$, the SHAP value ϕ_i of feature i for input $\mathbf{x}$ is the Shapley value from cooperative game theory:

$$\phi_i(f, \mathbf{x}) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} \times [f_{S \cup \{i\}}(\mathbf{x}) - f_S(\mathbf{x})] \quad (5)$$

where $f_S(\mathbf{x})$ is the expected model output over the marginal distribution of features not in S . $\phi_i > 0$ indicates that feature i pushes the prediction toward Attack; $\phi_i < 0$ pushes toward Natural. TreeSHAP [13] computes exact Shapley values for tree ensembles in $\mathcal{O}(TL^2d)$ time. AYLA selects the top- k features with $\phi_i > 0$ (alert drivers) and uses their names—after semantic expansion—as the primary RAG query. SHAP is applied to the LightGBM base model on the 29-feature input; contributions are zero-padded back to the full 128-feature space for display.

Cross-Encoder Reranking

Given primary query q and retrieved document d_j , the cross-encoder f_{CE} scores each pair jointly by concatenating the full text of q and d_j through a transformer:

$$s_j = f_{\text{CE}}(q, d_j) \quad (6)$$

Unlike bi-encoder L2 scores, cross-encoder scores are directly comparable across all candidates regardless of which retrieval pass found them. When frequency sensors appear in the alert drivers, a supplementary query q_{freq} is constructed and scores are summed:

$$\hat{s}_j = f_{\text{CE}}(q, d_j) + f_{\text{CE}}(q_{\text{freq}}, d_j) \quad (7)$$

Evaluation Metrics

The following metrics are reported for the binary classifier:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (8)$$

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (9)$$

where p_o is observed agreement and p_e is expected chance agreement. MCC is preferred over accuracy for imbalanced datasets as it accounts for all quadrants of the confusion matrix.

All proportions are reported with Wilson score 95% confidence intervals:

$$\tilde{p} \pm \Delta = \frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}} \quad (10)$$

with $z = 1.96$ for 95% coverage, n the sample size, and $\hat{p} = k/n$ the observed proportion.

C Implementation Details

Data preprocessing (preprocess_raw_data.py). Raw sensor data is stored as a collection of CSV files in the `binaryAllNaturalPlusNormalVsAttacks/` directory. The preprocessing script concatenates all files, strips whitespace from column names, and handles both `marker` and `@marker` target column variants across dataset releases. Feature columns are cast to numeric with `errors='coerce'`, $\pm\infty$ values are replaced with `NaN`, and all remaining `NaN` entries are filled with per-column medians. The cleaned dataframe is persisted as `processed_power_data.pkl` via `pandas.to_pickle` for downstream use.

Classifier training (train_classifier_v2.py and model_utils.py). LightGBM rejects feature names containing colons, so all column names are sanitized with a global `str.replace(':', '_')` before training. A

64/16/20 stratified split produces training, calibration, and test sets. Training proceeds in four stages: (A) an initial LightGBM model is trained on all 128 features and global SHAP importance is computed on a 3000-sample subset to select 29 above-mean-importance features; (B) LightGBM (600 estimators, 255 leaves) and XGBoost (600 estimators, depth 8) are retrained on the 29-feature subset; (C) a LogisticRegression meta-learner is fit on the calibration set’s stacked $[P_{\text{LGB}}, P_{\text{XGB}}]$ predictions; (D) a threshold sweep selects $\tau = 0.560$ as the accuracy-maximising threshold subject to false alert rate $< 15\%$. The resulting `EnsembleCalibratedClassifier` wrapper in `model_utils.py` is fully picklable by `joblib` from any entry point.

Protocol knowledge base (`build_rag_index.py`). The thirteen operator protocols are stored in a single structured plain-text file (`manuals/grid_protocols.txt`), each prefixed with a `PROTOCOL <ID>`: header line. The indexing script splits documents into 700-character chunks with 100-character overlap using `LangChain’s RecursiveCharacterTextSplitter`. Chunk size was chosen empirically: 700 characters keeps R3-FDIA and R4-PM (which together span ~ 790 characters) in separate chunks, preventing cross-protocol content bleeding, while still fitting each FDIA protocol in a single retrievable unit. Relay metadata is assigned by matching the `PROTOCOL R[1-4]` regex against each chunk’s content; chunks without a relay-specific header are tagged `GENERIC`. This header-based tagging was preferred over content scanning because multi-relay protocols such as `FREQ-01` and `EXPECTED-09` mention all four relay identifiers, and content scanning would incorrectly tag them as R1. Chunks are embedded with the `all-MiniLM-L6-v2` sentence encoder (384-dimensional output) and stored in a ChromaDB collection at `./chroma_db/` with L2 distance as the similarity metric.

Real-time monitor (`grid_monitor.py`). The production monitor loads the calibrated model, label encoder, and full dataset at startup, then iterates over randomly shuffled rows to simulate an event stream. A 12-entry regex registry (`_SENSOR_PATTERNS`) maps sensor name prefixes to human-readable type labels (e.g., `R\d-PM[1-3]:V` $\rightarrow$ “voltage magnitude”), enabling sensor names to be expanded into semantically richer query strings before embedding. LightGBM rejects colon characters in feature names, so column names are sanitized at load time; a reverse map (`_col_sanitized_to_original`) restores original colon-containing names on SHAP contributions so that the regex registry can match them correctly. When a sample exceeds the alert threshold τ , the monitor: (1) computes exact SHAP values via `shap.TreeExplainer` on the LightGBM base model; (2) zero-pads selected-feature SHAP values back to 128 dimensions; (3) selects the top- k positive contributions as alert drivers; (4) calls `get_smart_context()` to perform two-phase retrieval with cross-encoder reranking; and (5) calls `consult_ai()` to

stream the LLM response token-by-token via the Ollama REST API (<http://localhost:11434/api/generate>). The LLM is invoked with `temperature=0.0` for deterministic output and a 120-second request timeout.

D Cascaded Per-Scenario Classifier

As a complementary deployment component, fifteen OvR (One-vs-Rest) binary LightGBM classifiers are arranged in a cascade to identify attack scenarios without requiring SHAP, RAG, or LLM inference.

Architecture

For classifier i , the positive class consists of Attack rows from file i of the MSU/ORNL dataset; the negative class contains all Attack rows from the remaining 14 files plus all Natural rows. Class imbalance ($\approx 20:1$ negative/positive) is handled by setting `scale_pos_weight =  $N_{\text{neg}}/N_{\text{pos}}$` per classifier. A threshold sweep selects the operating point τ_i that maximises F1 on the test set for each classifier independently. Classifiers are arranged in a cascade ordered by descending validation F1; the first classifier to exceed τ_i determines the predicted scenario. If no classifier fires, the sample is predicted as Natural.

Table 9: Cascaded per-scenario classifier results on the 15,676-sample test set. “Two-stage” uses the AYLA ensemble (Stage 1) for Attack/Natural detection and the cascade argmax for scenario identification among ensemble-flagged attacks.

Configuration	Scen. Acc.	Bin. Acc.
OvR cascade (standalone)	81.1%	84.0%
OvR cascade (two-stage)	83.6%	95.8%
Multi-class LGB (ref)	83.6%	91.6%
AYLA ensemble (binary)	N/A	91.9%

Deployment Value

The cascade’s operational value is speed: scenario identification requires no SHAP computation, RAG retrieval, or LLM inference—it runs in under 1 ms per sample. In a two-stage deployment, the cascade provides immediate triage for 83.6% of events while the remaining 16.4% are escalated to the full AYLA SHAP+RAG+LLM pipeline for high-confidence attribution. This reduces mean alert response latency from the AYLA pipeline’s approximately 3.5 minutes per event to approximately 2.5–5 seconds for cascade-triaged events ($0.164 \times 3.5 \text{ min}$).

E Threshold Sweep Analysis

The optimal threshold $\tau=0.560$ is selected by sweeping $[0.30, 0.90]$ in steps of 0.005 and maximising accuracy sub-

Table 10: Threshold sweep showing the precision-recall trade-off. Optimal $\tau=0.560$ maximises accuracy subject to false alert $< 15\%$.

Threshold	Atk Recall	Nat Recall	False Alert%	F1-wtd
0.30	97.0%	78.5%	21.5%	0.9144
0.40	96.1%	81.4%	18.6%	0.9174
0.50	95.2%	84.0%	16.0%	0.9187
0.560	94.6%	85.3%	14.7%	0.9186
0.60	94.1%	86.0%	14.0%	0.9174
0.65	93.3%	86.9%	13.1%	0.9148
0.70	92.5%	88.1%	11.9%	0.9131

ject to the constraint that the false alert rate (fraction of Natural samples misclassified as Attack) remains below 15%. Table 10 shows representative thresholds across this range. As τ increases, attack recall decreases while natural recall (and the operator experience) improves; the optimal point balances these opposing objectives under the false-alert constraint.

F Additional Failure Mode Analysis

CURR-01 and DEGR-02. CURR-01’s 1/4 accuracy and DEGR-02’s 1/2 accuracy reflect their small sample counts (4 and 2, respectively); additional evaluation samples would clarify whether these are systematic failure modes or statistical noise.

PHASE-01 cross-attribution bias. The persistent PHASE-01 bias (3 false citations across 200 samples) reflects a document-level issue: PHASE-01’s protocol text contains broad language about “voltage anomalies” and “sensor magnitude deviations” that overlaps with FDIA attack descriptions. Tightening PHASE-01’s language to emphasize its unique discriminator (simultaneous step-change in phase angle *with stable magnitudes*) would reduce its embedding similarity to voltage-magnitude queries.

Hardware deployment detail. Full throughput estimates by GPU class:

- **NVIDIA RTX 4090** (24 GB VRAM, $\sim 1,008$ GB/s bandwidth, $\sim 3\times$ the laptop 3060): estimated **45–90 s** per pipeline event, enabling FP16 inference instead of INT4 quantization; approximately **40–80 flagged events per hour**.
- **NVIDIA A100** (80 GB VRAM, $\sim 2,039$ GB/s bandwidth, $\sim 6\times$ improvement): estimated **20–40 s** per event, approximately **90–180 events per hour**.
- **Multi-substation deployment:** a clustered GPU server with $2\text{--}4\times$ A100s provides real-time concurrent coverage across multiple relay systems.